

Daily Operations & Site Progress Report

Mucheru World of Golf dam earthworks & green construction, Chaka Farms veterinary care & canine training, Kabete Residence tank slab & fireplace demolition, Amani Cottage grass planting, and DeKUT collaboration visit.

TOTAL HAULAGE
TRIPS

82 Trips

Murram & excavated soil
haulage

DAM 2 EXCAVATION

60% Complete

Excavator 9h 51m |
Backhoe 6h 53m

MURRAM DELIVERIES

5 Lorries

From Dam 1 to clubhouse
ground

CHAKA MILK YIELD

7 Litres

Morning: 4L | Evening: 3L

1. Mucheru World of Golf — Dam Works, Clubhouse



Profiling & Greens 1-3

Front loader & backhoe earthmoving, Dam 1 murram haulage, Dam 2 excavation, green foundation work, PVC irrigation prep, and greenhouse wind damage

Clubhouse Soil Profile & Dam 1 Levelling

Overseen by: Ken & Mr. Gichuhi

- Front Loader Arrival:** The front loader arrived on site at **10:00 AM** and immediately commenced digging the **soil profile hole for the clubhouse** foundation assessment.
- Dam 1 Levelling & Murram Collection:** The front loader continued levelling **Dam 1**, gathering murram from the basin for redistribution across the project.
- Murram Haulage:** **Five lorry loads** of murram excavated from Dam 1 were transported to the ground area near the clubhouse for stockpiling and future use.
- Backhoe Dam Levelling:** A yellow XGMA backhoe loader was deployed to level the dam basin floor, leaving parallel bucket-grade lines across the excavated surface.

Dam 2 Excavation, Soil Dumping & Foundation Breaking

Overseen by: [Ken & Mr. Gichuhi](#)

- **Dam 2 Excavation Progress:** Excavation of the **2nd Dam** reached approximately **60% completion**, with a Sunward excavator and tipper trucks operating across the basin.
- **Soil Dumping & Haulage:** Excavated spoil was loaded onto lorries and dumped across designated low-lying ground to establish smooth contours — totalling **82 haulage trips** for the day.
- **Foundation Breaking:** Crews broke through existing foundation layers within the dam excavation zone to reach the required murram stratum.
- **Active Dam Digging:** Continuous excavator bucket work deepened the basin walls, exposing distinct red-soil and murram geological layers.

EQUIPMENT	OPERATING HOURS	PRIMARY TASK
Excavator	9 Hours 51 Minutes	Dam 2 digging, soil loading & dumping
Backhoe Loader	6 Hours 53 Minutes	Dam levelling, murram gathering & profiling
Total Haulage Trips	82 Trips	Murram & excavated soil transport

 Dam 2 Excavation Progress

[View on Google Drive ↗](#)

Greens 1, 2 & 3 Construction & Irrigation Prep

Overseen by: [Ken & Mr. Gichuhi](#)

- **Green Foundation Work:** Construction progress continued on **Greens 1, 2, and 3**, including sub-base preparation and material placement.
- **Red Soil Levelling — Green 3:** Crews levelled red topsoil across **Green Number 3** to establish the growing surface profile.
- **PVC Pipe Measurements & Slitting:** Orange PVC irrigation pipes were measured, cut, and slit at regular intervals using an angle grinder in preparation for sub-surface drainage installation.

 Greens 5 & 7 Turf Growth

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Greenhouse Wind Damage

- **Wind Damage Incident:** Strong winds tore through the greenhouse plastic covering, ripping large sections from the metal hoop frame and exposing the interior to the elements.

⚠ Greenhouse Covering Requires Repair

Multiple large tears and detached plastic sheets were observed on the greenhouse roof and sidewalls. Repair or replacement of the covering material is needed before the next weather event.



Mucheru World of Golf Visual Documentation



Clubhouse Soil Profile Test Pit

Deep vertical trench excavated by front loader showing clean-cut soil layers for clubhouse foundation profiling.



XGMA Backhoe Loader — Site Standby

Yellow XGMA 765N backhoe loader with front bucket lowered and rear arm folded, parked on dry grass near tree line.



Backhoe Levelling Dam Basin Floor

XGMA backhoe actively grading the wide dam excavation floor, leaving parallel horizontal bucket marks across the surface.



Backhoe Active Levelling Inside Dam

Backhoe loader pushing and levelling murrum inside the dam bowl; tire tracks and dust visible on the basin floor.



Dam 2 Excavation — Excavator & Lorries

Sunward teal excavator loading soil with white, yellow, and blue tipper trucks lined up at the Dam 2 haulage zone.



Dam 2 Basin — End-of-Day Overview

Broad perspective of the excavated Dam 2 reservoir showing sloped embankments and spoil mound at twilight.



Dam 2 Red-Soil Wall Excavation

Steep tiered dam walls exposing red murram strata; perimeter fence and white pickup visible on the access road above.



PVC Pipes Measured & Slitted

Bundle of orange PVC irrigation pipes with uniform longitudinal slits cut at regular intervals for drainage installation.



PVC Pipe Slitting with Angle Grinder

Worker in blue overalls cutting slits into long orange PVC pipe using a handheld angle grinder beside a stacked pipe bundle.



Greenhouse Roof — Wind Tear Damage

Large sections of white greenhouse plastic torn from the central peak, flapping open and exposing the metal hoop frame to sky.



Greenhouse Sidewall — Wind Shredding

Close-up of shredded greenhouse sidewall plastic with jagged tears; cypress tree line visible through the damaged opening.

2. Chaka Farms — Veterinary Care, Dairy, Grounds & Canine Training



Veterinary injections & deworming, dairy milking logs, crop & flower irrigation, compound maintenance, and canine obedience training



Veterinary Visit — Injections & Deworming

Overseen by: [Kamandau](#)

- **Veterinary Attendance:** A veterinarian visited Chaka Farms to administer **injections and deworming treatment** to the goats and cattle herd.
- **Selective Deworming:** Deworming was specifically administered to cows that are **not in the milking rotation**, ensuring treatment without affecting dairy production schedules.



Dairy Production & Milking Yield

Overseen by: [Kamandau](#)

SESSION	MORNING	EVENING	TOTAL DAILY YIELD
Milk Yield	4 Litres	3 Litres	7 Litres



Flower, Grass & Crop Irrigation

Overseen by: [Kinoti](#)

- **Flower & Planted Grass Watering:** Flowers and newly planted grass areas were irrigated to maintain moisture levels during the dry season.
- **Crop Watering:** Field crops received scheduled watering to support growth through the dry period.

Compound Cleanliness, Gardening & Grass Maintenance

Overseen by: **Monica & Margaret**

- **General Compound Cleanliness:** Routine sweeping, litter collection, and tidying across farm compound areas.
- **Gardening:** Garden bed maintenance including weeding and plant care across landscaped zones.
- **Compound Grass Maintenance:** Lawn mowing, edging, and general grass upkeep on compound turf areas.

Canine Unit — Manners, Patience & Goat Socialization

Overseen by: **Willy**

- **Kennel Manners & Patience:** Structured training sessions focused on calm kennel behaviour, waiting commands, and impulse control.

 Kennel Manners & Patience

[View on Google Drive ↗](#)

- **Off-Leash Walking & Goat Socialization:** Dogs exercised off-leash alongside grazing goats, building social tolerance and controlled interaction during pasture walks.

 Off-Leash Walking & Goat Socialization — Clip 1

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 Off-Leash Walking & Goat Socialization — Clip 2

[View on Google Drive ↗](#)

 Off-Leash Walking & Goat Socialization — Clip 3

[View on Google Drive ↗](#)

- **Patience & Obedience Drills:** Reinforcement of sit, stay, and patience commands to improve overall obedience and handler responsiveness.

 Patience & Obedience Training

[View on Google Drive ↗](#)

Office & Administrative Notes

- **Office TV Collection:** Mr. Maina collected the office television for professional framing.
- **Staff Reporting:** Jackie Kamau reported on duty as assistant to Gladys.



Veterinary Injection — Calf Treatment

Vet in grey jacket administering injection to a dark-brown calf restrained by a Chaka Farms worker in green coveralls.



Calf in Treatment Pen

Black-and-white calf standing in a rustic wooden livestock pen following veterinary attendance.



Calf Deworming Session

Light-tan calf tethered to a green post while vet in blue hooded jacket and rubber boots attends in the pen.



Planted Grass Irrigation

Water stream from sprinkler soaking a straw-mulched grass field with emerging green patches under cypress tree line.



Crop & Seedling Watering

Green garden hose watering young plants on straw-mulched reddish soil with tilled field and grazing cattle in background.



Office TV Collected for Framing

Mr. Maina and assistant loading the office television into a red car parked on the driveway for professional framing.

3. Kabete Residence — Painting, Tank Slab & Fireplace Demolition



Main house shelf painting, downstairs grill paintwork, water storage tank slab pour, outdoor fireplace demolition, and domestic pet care



Interior & Exterior Paint Works

- **Main House Shelf Painted:** A worker on a ladder completed painting the main house shelf area above the stone-walled wooden door entrance.
- **Downstairs Grill Painting:** Black metal handrail and grillwork alongside the outdoor stone staircase received a fresh coat of paint, with protective cloth laid on the steps.



Main House Water Storage Tank Slab

- **Concrete Mixing & Slab Pour:** Masonry crew manually mixed ballast, cement, and sand on site and began laying the **main house water storage tank foundation slab**.
- **Slab Compaction:** A concrete vibrator was used to compact and firm the freshly poured wet slab surface for structural integrity.



Tank Slab Pour Underway

Main house storage tank foundation slab mixing, pouring, and vibration compaction actively progressing on site.



Outdoor Fireplace Demolition

- **Fireplace Structure Demolition:** Workers dismantled the circular outdoor fireplace seating wall using sledgehammers and crowbars, breaking down the stone-clad masonry structure.
- **Foundation Excavation:** Red soil was excavated from within the circular fireplace pit to clear the site for future reconstruction work.



Domestic Pet Care (Kabete Residence)

- **Evening Feeding:** Golden retriever *Ajabu* and resident cats received their evening meals on the terracotta tiled veranda and interior floor.



Kabete Residence Visual Documentation



Main House Shelf & Overhang Painting

Worker in red paint-splattered coveralls on ladder painting the tiled overhang and shelf area above the stone-walled wooden door.



Downstairs Grill & Handrail Painting

Worker in red overalls painting black metal handrail and grill alongside stone steps, with white protective cloth on stair treads.



Tank Slab — Concrete Mixing

Four workers in safety vests manually mixing ballast and cement with shovels beside Blue Triangle cement bags near stone wall foundation.



Slab Vibration & Compaction

Worker in tan coveralls squatting over freshly poured wet concrete, using a vibrator to compact the tank slab surface.



Fireplace Demolition — Pit Excavation

Worker in yellow safety vest inspecting circular excavated pit inside the curved stone-clad fireplace seating structure.



Fireplace Wall Breaking

Two workers in safety vests dismantling the curved fireplace masonry wall with sledgehammer and crowbar; rubble piled inside.



Fireplace Demolition — Foundation Cleared

Four-man crew working in the red-soil excavated fireplace pit with sledgehammer and scattered grey stone blocks.



Ajabu — Evening Dinner

Golden retriever Ajabu eating kibble from a bowl on the terracotta tiled veranda in afternoon sunlight.



Resident Cats — Evening Dinner

Three cats — white-tabby, calico, and tortoiseshell — eating simultaneously from stainless steel bowls on tiled floor.



4. Amani Cottage — Grass Planting & Irrigation

New grass planting and sprinkler irrigation across cottage grounds

Overseen by: [Edwin](#)

 Grass Planting & Irrigation

- **Grass Planting:** New grass tufts were planted across prepared dark soil beds in the cottage garden area.
- **Immediate Irrigation:** Planted grass areas were thoroughly irrigated using a garden hose and sprinkler to establish root moisture.

 Amani Cottage Visual Documentation



New Grass Planted

Small green grass tufts evenly spaced across dark prepared soil bed bordered by trees and rock-edged garden bed.



Planted Grass Irrigation

Rotary sprinkler actively watering newly planted grass patches with green garden hose routed across the lawn area.



5. DeKUT Collaboration Visit — DSAIL Lab

Presentations

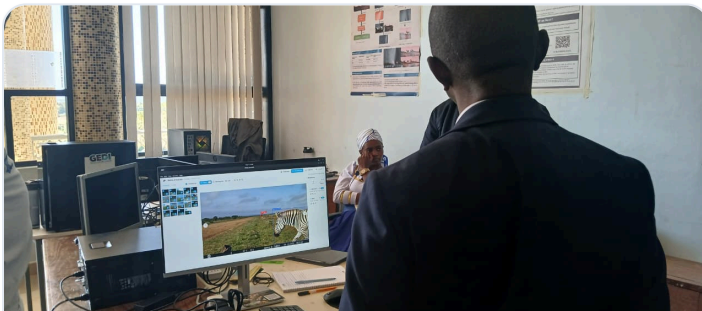
Meeting with Vice Chancellor & Prof. Ciira; postgraduate research presentations at the DSAIL Lab

📍 University Visit & Research Collaboration

- **DeKUT Visit:** The team visited **Dedan Kimathi University of Technology (DeKUT)**, meeting with the **Vice Chancellor** and **Prof. Ciira**.
- **DSAIL Lab Tour:** Postgraduate researchers at the **DSAIL (Data Science & AI Lab)** presented ongoing projects including water level data analysis, crop monitoring, computer vision for wildlife tracking, speech recognition for African languages, and medical imaging research.
- **Collaboration Exploration:** The visit provided an opportunity to explore potential areas of collaboration between DeKUT research expertise and Mr. Mucheru's ongoing agricultural and infrastructure projects.

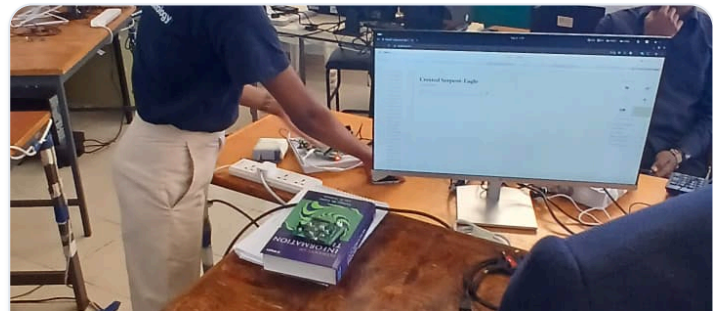


DeKUT DSAIL Lab Visual Documentation



Wildlife Computer Vision Demo

DSAIL researcher demonstrating zebra object-detection software with bounding boxes on a lab monitor; research posters on wall.



Researcher Project Presentation

Postgraduate researcher presenting at a DSAIL workstation; Data Science Africa (DSA) banner visible in background.



Water Level Data Analysis — Crop Monitoring

Presentation slide showing water level data analysis graphs for crop monitoring; visiting team observing at DSAIL lab desk.



Hardware & Sensor Discussion

Research team discussing electronic sensor hardware and circuit boards at DSAIL lab bench with "Elements of Information Theory" textbook.



Tunuh Speech Recognition System

Presentation of "Tunuh: Automatic Speech Aware System for Low-Resource African Languages" by researcher Maly Jorku at DSAIL.



Medical Imaging & Crown Growth Tracking

Dual-monitor presentation showing lower limb alignment measurements and satellite-based crown growth tracking research projects.

DN

Compiled by Dennis Ndirangu